E-Commerce Marketing Performance Analytics

Revenue Growth, Channel Efficiency & Customer Retention Strategy

A data analytics project analyzing e-commerce business performance through revenue trends, marketing channel efficiency, customer acquisition, retention behavior, and product category insights using Python analytics and Power BI dashboard visualization.

1. Executive Summary

This project presents a comprehensive analysis of an e-commerce business performance dataset to identify the major factors affecting revenue growth, marketing efficiency, customer retention, and category performance.

The analysis combines business metrics, marketing KPIs, and customer behavior indicators to evaluate whether current growth strategies are sustainable. The study focuses on revenue trends, advertising efficiency, channel performance, customer retention, and product category contribution.

2. Project Objective

The main objective of this project is to transform raw business data into actionable insights that support strategic decision-making.

The analysis aims to:
• Identify reasons behind slowing revenue growth.
• Evaluate marketing spending efficiency.
• Compare performance across acquisition channels.
• Analyze customer retention behavior.
• Identify high-performing product categories.
• Recommend strategies for sustainable growth.

3. Data Analysis Approach

The project was developed using a structured analytics workflow:

1. Data preparation and cleaning
2. Monthly business performance analysis
3. Marketing efficiency evaluation
4. Customer retention analysis
5. Category performance analysis
6. Strategic recommendation development

The analysis used Python-based data analysis and visualization, while dashboard visualization was developed separately using Power BI.

4. Revenue Growth Analysis

The analysis reveals a gap between marketing investment growth and revenue growth.

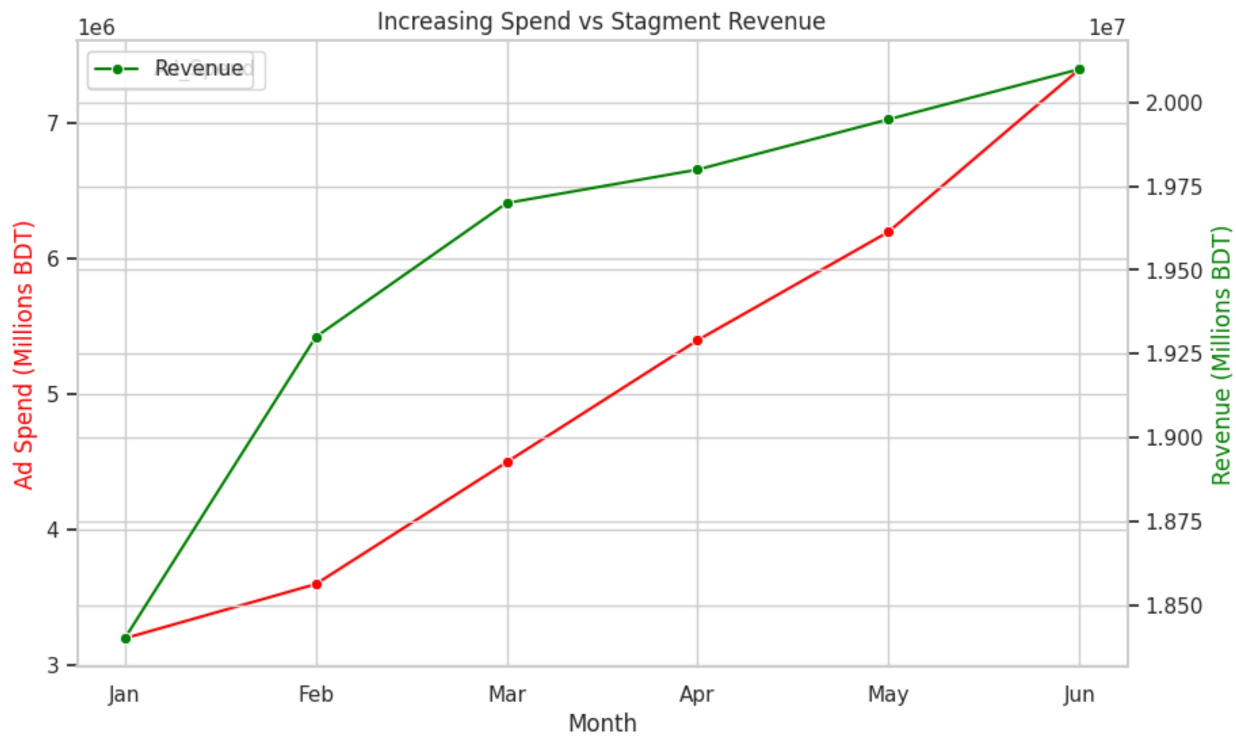

Figure: Increasing Spend vs Stagment Revenue

Key findings:

• Revenue increased approximately 9.2% over the six-month period.
• Advertising spends increased approximately 131.2%.
• The gap indicates diminishing returns from additional marketing investment.

The business is spending significantly more to generate relatively smaller revenue growth, suggesting that scaling acquisition without improving efficiency is not sustainable.

5. Conversion Rate & Traffic Quality Analysis

Traffic volume increased significantly; however, traffic quality declined.

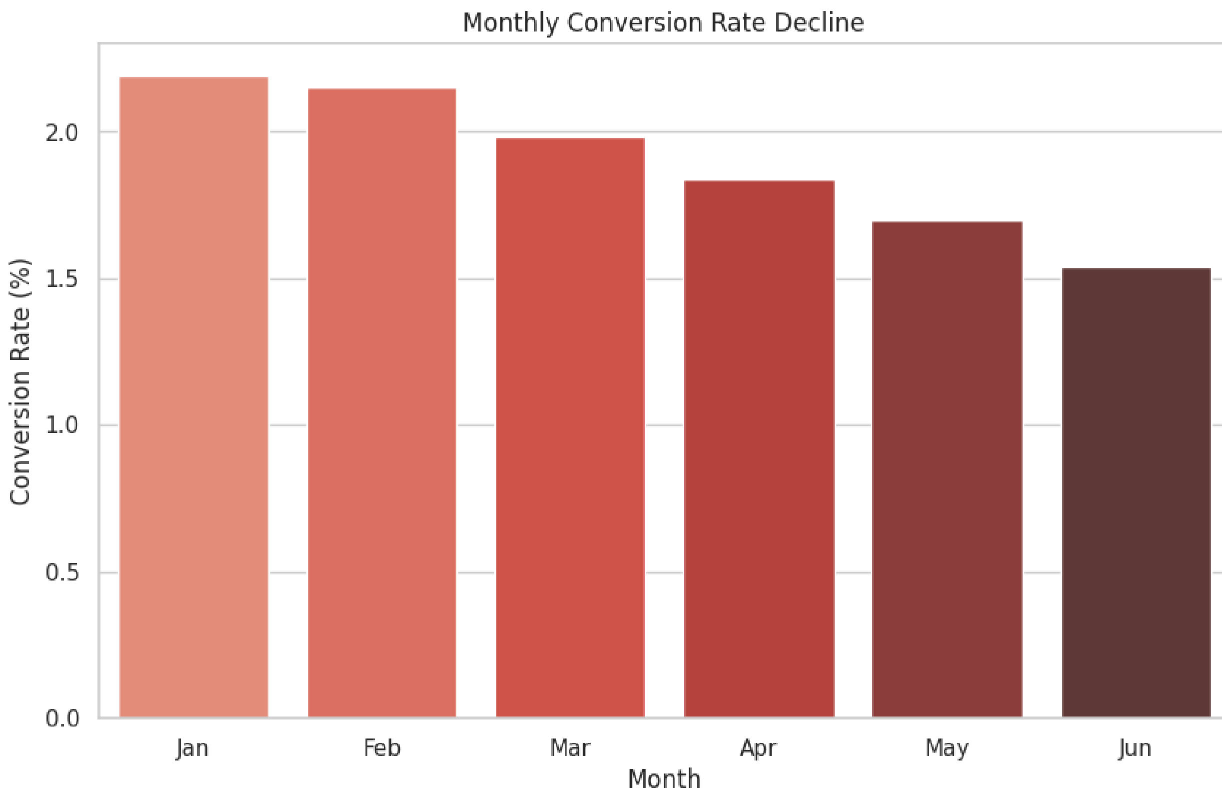

Figure: Monthly Conversion Rate Decline

Findings:

• Sessions increased from 420,000 to 680,000 (+62%).
• Orders increased only from 9,200 to 10,500 (+14%).
• Conversion rate declined from 2.19% in January to 1.54% in June.

This indicates that the company is attracting more visitors but a lower percentage of high-intent customers.

6. Marketing Channel Performance Analysis

Channel-level analysis was performed using key marketing KPIs:

• Customer Acquisition Cost (CAC)
• Return on Ad Spend (ROAS)
• Conversion Rate

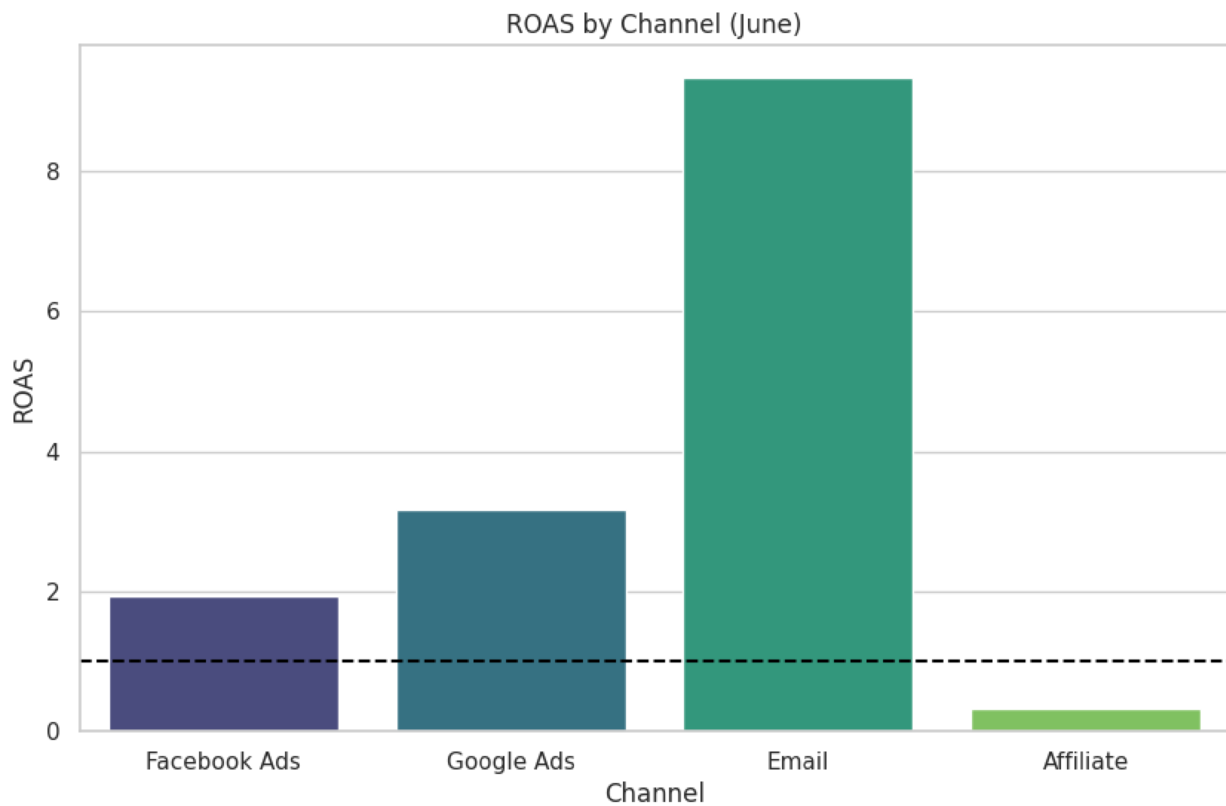

Figure: ROAS by Channel

Key insights:

Google Ads:
- Strong paid acquisition performance.
- Conversion rate around 2.05%.
- CAC approximately 641 BDT.

Email Marketing:
- Highest efficiency channel.
- ROAS approximately 9.33.
- Indicates strong potential from existing customer database.

Affiliate Marketing:
- Weakest performing channel.
- CAC approximately 2,250 BDT.
- ROAS approximately 0.33, indicating inefficient spending.

7. Customer Retention Analysis

The analysis identifies a customer retention challenge.

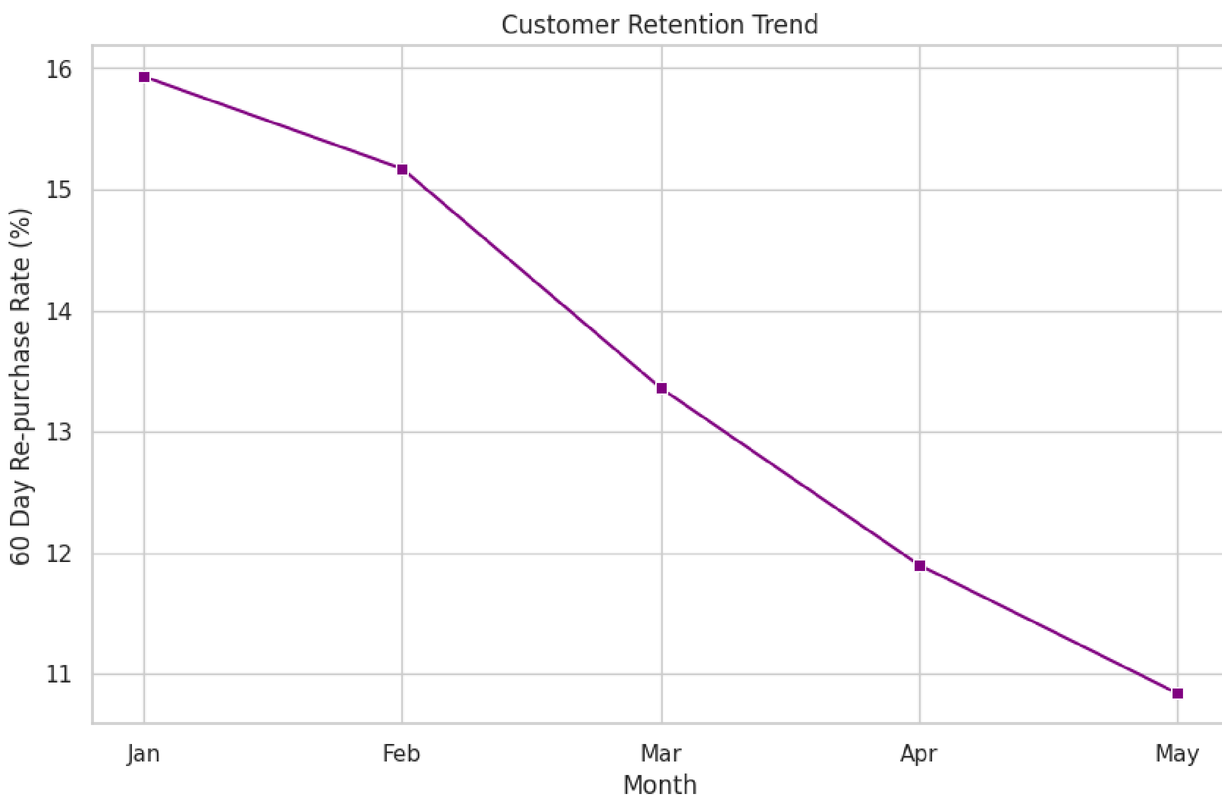

Figure: customer retention trend

Findings:

• New customer acquisition increased significantly.
• 60-day repeat purchase rate declined from 15.9% in January to 10.8% in May.

This creates a 'leaky bucket' situation where the company invests heavily in acquiring customers but loses long-term customer value due to weak retention.

Strategic focus should shift from only acquisition toward loyalty-building activities.

8. Category Performance Analysis

Product category performance was analyzed based on orders, revenue, and average order value.

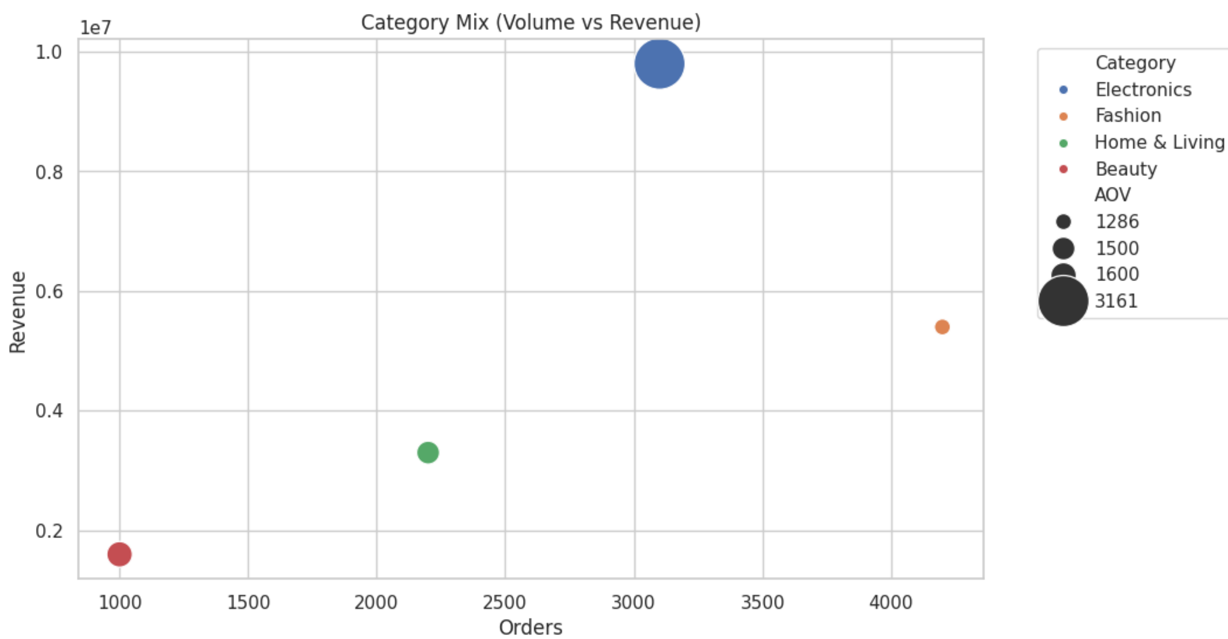

Figure: Category Mix Compare

Key findings:

Fashion:
- Highest order volume.
- 4,200 monthly orders.
- Strong customer acquisition potential.

Electronics:
- Highest revenue contributor.
- Generated 9,800,000 BDT.
- Nearly 49% of total monthly revenue.
- Highest Average Order Value (AOV): 3,161 BDT.

Strategic implication:
Use Fashion for customer acquisition and Electronics for revenue growth.

9. Strategic Recommendations

Improve Marketing Efficiency
Optimize advertising budget allocation toward channels with stronger ROAS and conversion performance.

Prioritize Retention
Develop CRM, email loyalty campaigns, and repeat purchase strategies.

Optimize Customer Quality
Focus on high-intent traffic instead of only increasing visitor volume.

Reduce Inefficient Channels
Review affiliate marketing strategy due to poor acquisition economics.

Strengthen High-Value Categories
Promote Electronics for revenue growth while using Fashion for customer acquisition.

Increase Average Order Value
Introduce product bundles and cross-category recommendations.

10. Project Evaluation

The analysis challenges the assumption that increasing traffic automatically indicates business health.

Traffic growth alone is a vanity metric. Sustainable growth requires:

• Revenue growth aligned with marketing expenses.
• Strong customer retention.
• Improving Customer Lifetime Value (LTV).
• Efficient Customer Acquisition Cost (CAC).

The business should prioritize quality over quantity by improving conversion efficiency and customer loyalty.

11. Tools & Technologies

- Python (Pandas, Matplotlib, Seaborn)
- Data analysis and KPI calculation
- Business visualization
- Power BI dashboard development
- Marketing performance analytics

12. Conclusion

This project demonstrates how business analytics can identify hidden growth problems behind increasing sales activity. The analysis shows that sustainable growth depends on balancing acquisition, conversion, retention, and profitability.

By reallocating marketing investment, improving customer loyalty, and focusing on high-value categories, the company can build a more efficient and sustainable growth model.